

Assignment No: – 7

ON

Full Stack Development – SSCA4020

AIM: Create A simple full website where client will send request, server handle and store data in the MongoDB database. Tech: React, Nodejs ,MongoDB..

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
(HONOURS.)

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CODE:

Commands

```
npm init -y
npm install express mongodb
node server.js
```

server.js

```
const express = require("express");
const { MongoClient, ObjectId } = require("mongodb");
const path = require("path");

const app = express();
const PORT = 4000;
const client = new MongoClient("mongodb://127.0.0.1:27017");
let tasks;

app.use(express.json());
app.use(express.static(path.join(__dirname, "public")));

// READ - all tasks
app.get("/api/tasks", async (req, res) => {
  res.json(await tasks.find().sort({ _id: 1 }).toArray());
});

// CREATE - new task
app.post("/api/tasks", async (req, res) => {
  const { title, priority } = req.body;
  if (!title || !title.trim()) return res.status(400).json({ message: "Title is required" });
  const task = { title: title.trim(), priority: priority || "Medium", done: false };
  const r = await tasks.insertOne(task);
  res.status(201).json({ ...task, _id: r.insertedId });
});

// UPDATE - toggle done / edit
app.put("/api/tasks/:id", async (req, res) => {
  await tasks.updateOne({ _id: new ObjectId(req.params.id) }, { $set: req.body });
  res.json({ message: "Updated" });
});

// DELETE
app.delete("/api/tasks/:id", async (req, res) => {
  await tasks.deleteOne({ _id: new ObjectId(req.params.id) });
  res.json({ message: "Deleted" });
});

client.connect().then(async () => {
  tasks = client.db("taskmanager").collection("tasks");
  await tasks.deleteMany({});
  console.log("MongoDB connected");
  app.listen(PORT, () => console.log(`Task Manager running at http://localhost:${PORT}`));
});
```

public/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Task Manager - Mini Project</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>Task Manager</h1>
    <p class="sub">Full Stack Mini Project &mdash; Node.js + Express + MongoDB</p>

    <div class="add">
      <input id="title" placeholder="Enter a new task...">
      <select id="priority">
        <option>High</option><option selected>Medium</option><option>Low</option>
      </select>
    </div>
  </div>
</body>
</html>
```

```

    </select>
    <button onclick="addTask()">Add Task</button>
  </div>
  <p id="msg" class="msg"></p>

  <div class="stats">
    <span id="total">Total: 0</span>
    <span id="pending">Pending: 0</span>
    <span id="completed">Completed: 0</span>
  </div>

  <ul id="list"></ul>
</div>
<script src="app.js"></script>
</body>
</html>

```

public/style.css

```

* { box-sizing: border-box; }
body { margin: 0; font-family: 'Segoe UI', Arial, sans-serif; background: linear-
gradient(135deg,#667eea,#764ba2); min-height: 100vh; padding: 30px 12px; }
.container { max-width: 640px; margin: auto; background: #fff; border-radius: 14px; padding: 28px; box-
shadow: 0 10px 30px rgba(0,0,0,.25); }
h1 { margin: 0; color: #3b3b98; } .sub { color: #777; margin-top: 4px; }
.add { display: flex; gap: 8px; margin-top: 18px; }
.add input { flex: 1; padding: 10px; border: 1px solid #bbb; border-radius: 8px; }
.add select { padding: 10px; border-radius: 8px; border: 1px solid #bbb; }
button { background: #3b3b98; color: #fff; border: 0; padding: 10px 16px; border-radius: 8px; cursor:
pointer; }
button:hover { background: #575fcf; }
.msg { color: #d63031; min-height: 18px; margin: 6px 0; font-size: 14px; }
.stats { display: flex; justify-content: space-between; background: #f1f2f6; padding: 10px 14px; border-
radius: 8px; font-weight: 600; font-size: 14px; }
ul { list-style: none; padding: 0; margin-top: 14px; }
li { display: flex; align-items: center; gap: 10px; padding: 12px; border-bottom: 1px solid #eee; }
li .t { flex: 1; cursor: pointer; } li.done .t { text-decoration: line-through; color: #999; }
li .badge { font-size: 12px; padding: 3px 10px; border-radius: 20px; color: #fff; }
li .High { background: #e74c3c; } .li .Medium { background: #f39c12; } .li .Low { background: #27ae60; }
li .del { background: #e74c3c; padding: 6px 12px; font-size: 13px; }

```

public/app.js

```

const API = "/api/tasks";

async function loadTasks() {
  const res = await fetch(API);
  const tasks = await res.json();
  const list = document.getElementById("list");
  list.innerHTML = "";

  tasks.forEach((t) => {
    const li = document.createElement("li");
    if (t.done) li.className = "done";
    li.innerHTML = `
      <input type="checkbox" ${t.done ? "checked" : ""} onchange="toggleTask('${t._id}', ${!t.done})">
      <span class="t" onclick="editTask('${t._id}', '${t.title.replace(/'/g, '\\')}')">${t.title}</span>
      <span class="badge ${t.priority}">${t.priority}</span>
      <button class="del" onclick="deleteTask('${t._id}')">Delete</button>;
    `;
    list.appendChild(li);
  });

  const done = tasks.filter((t) => t.done).length;
  document.getElementById("total").textContent = "Total: " + tasks.length;
  document.getElementById("pending").textContent = "Pending: " + (tasks.length - done);
  document.getElementById("completed").textContent = "Completed: " + done;
}

async function addTask() {
  const title = document.getElementById("title").value;
  const priority = document.getElementById("priority").value;
  const res = await fetch(API, {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ title, priority })
  });
}

```



```
});  
const msg = document.getElementById("msg");  
if (!res.ok) { msg.textContent = (await res.json()).message; return; }  
msg.textContent = "";  
document.getElementById("title").value = "";  
loadTasks();  
}  
  
async function toggleTask(id, done) {  
  await fetch(`${API}/${id}`, { method: "PUT", headers: { "Content-Type": "application/json" }, body:  
JSON.stringify({ done }) });  
  loadTasks();  
}  
  
async function editTask(id, oldTitle) {  
  const title = prompt("Edit task", oldTitle);  
  if (title && title.trim()) {  
    await fetch(`${API}/${id}`, { method: "PUT", headers: { "Content-Type": "application/json" }, body:  
JSON.stringify({ title: title.trim() }) });  
    loadTasks();  
  }  
}  
  
async function deleteTask(id) {  
  await fetch(`${API}/${id}`, { method: "DELETE" });  
  loadTasks();  
}  
  
loadTasks();
```

OUTPUT:

```
Windows PowerShell - node server.js  
  
PS C:\Users\Rishabh\FSD\mini-project> npm init -y  
PS C:\Users\Rishabh\FSD\mini-project> npm install express mongodb  
added 70 packages, and audited 71 packages in 5s  
  
found 0 vulnerabilities  
PS C:\Users\Rishabh\FSD\mini-project> node server.js  
MongoDB connected  
Task Manager running at http://localhost:4000
```

Task Manager - Mini Project

http://localhost:4000



Task Manager

Full Stack Mini Project — Node.js + Express + MongoDB

Enter a new task... Medium Add Task

Total: 0 **Pending: 0** **Completed: 0**

Task Manager - Mini Project

http://localhost:4000



Task Manager

Full Stack Mini Project — Node.js + Express + MongoDB

Enter a new task... High Add Task

Total: 4 **Pending: 4** **Completed: 0**

- ☐ Complete FSD lab manual High Delete
- ☐ Revise React hooks Medium Delete
- ☐ Push project to GitHub Low Delete
- ☐ Prepare for viva High Delete

Task Manager - Mini Project

http://localhost:4000



Task Manager

Full Stack Mini Project — Node.js + Express + MongoDB

Enter a new task... Medium Add Task

Total: 3 **Pending: 2** **Completed: 1**

- ☒ Complete-FSD-lab-manual High Delete
- ☐ Revise React hooks and Express routes Medium Delete
- ☐ Prepare for viva High Delete